

Cassie_Duncan_Foundations_in_Data_Science_Final_Project

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First Check-In

Generate Summary Statistics

```
# read in the Sample RNA-seq Count Matrix
gene_df <- read.csv("/Users/cassieschool/Documents/Foundations in Data Science/Final Project/Data/count.

# Convert the data frame to long format from wide format
gene_df_long <- tidyr::pivot_longer(gene_df, cols = -X, names_to = "barcode", values_to = "Count")

# Subset data to only include one selected gene of interest (TACSTD2)
one_gene <- gene_df_long[gene_df_long$X == "ENSG00000184292.7", ]

# Generate full summary statistics (minimum, first quantile, median, mean, third quantile, and maximum)
summary(one_gene$Count)
```

```
##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##         1     9866   16523   19879   25916  119596
```

```
# Generate Standard Deviation separately since it is not included in the Summary Statistics
cat("Standard Deviation", "\n", sd(one_gene$Count))
```

```
## Standard Deviation
##  15161.47
```

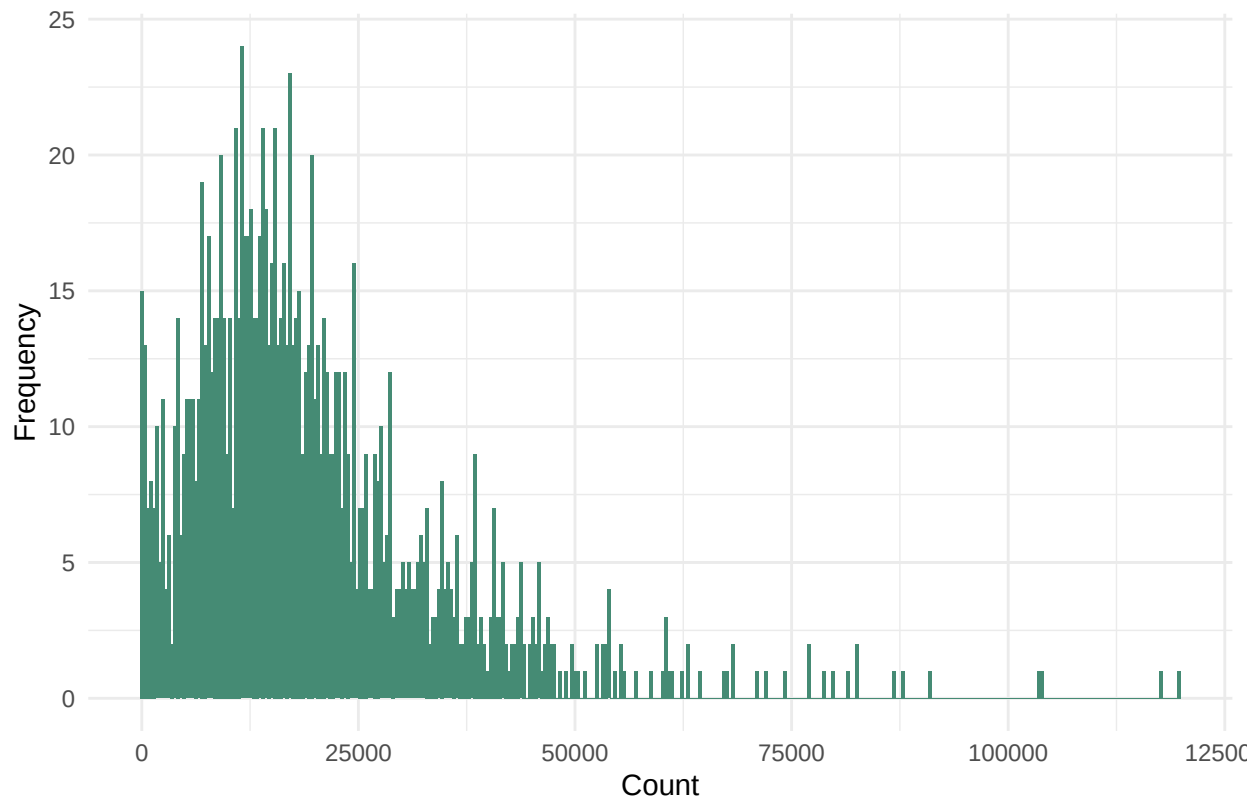
Create Histogram

```
library(ggplot2)

# Create histogram of one gene of interest
gene_plot <- ggplot(one_gene, aes(x = Count)) +
  geom_histogram(binwidth = 350, bins = 9000, # set bins and bin width
                 fill = "aquamarine4") + #choose color of plot
  # add titles to plot and axes
  labs(title = "Distribution of Counts for Gene TACSTD2",
        x = "Count",
        y = "Frequency") +
  theme_minimal()

print(gene_plot)
```

Distribution of Counts for Gene TACSTD2



Create Scatter Plot

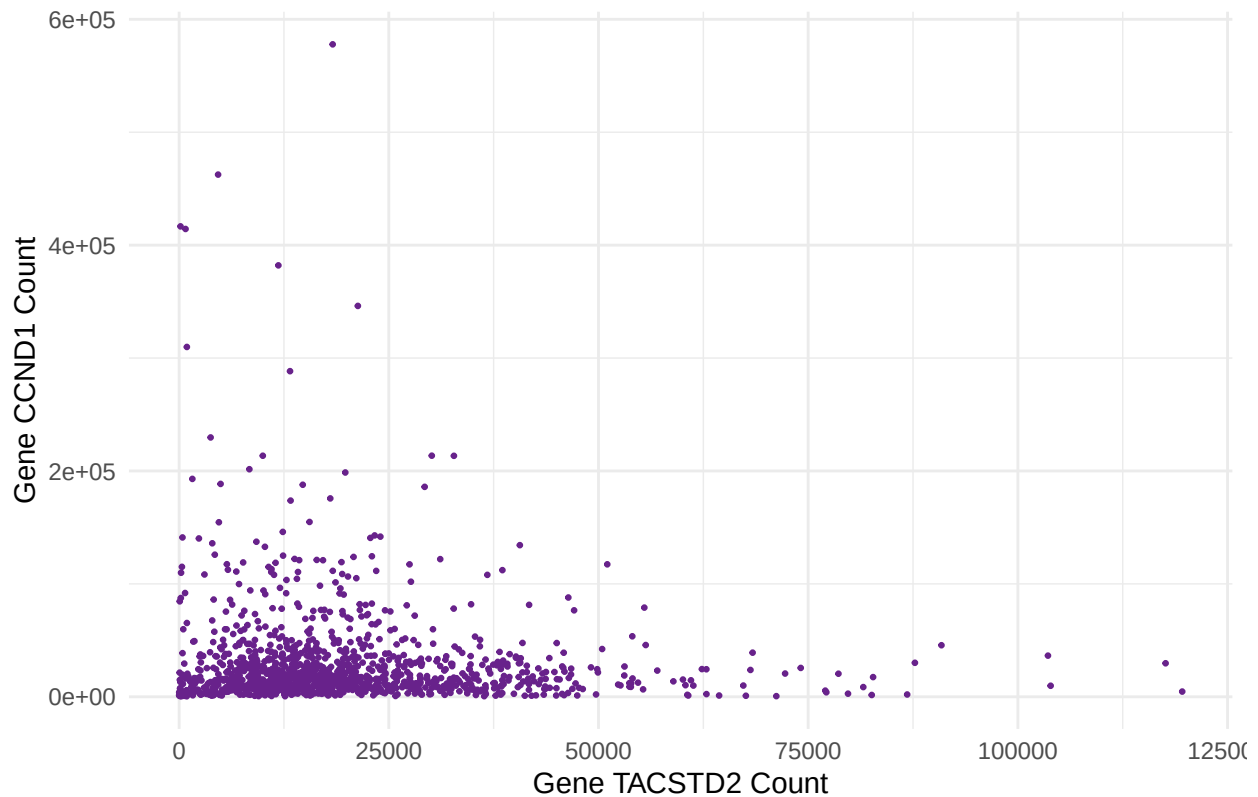
```
# subset data for only two genes of interest
two_genes <- gene_df_long[gene_df_long$X == "ENSG00000184292.7" | gene_df_long$X == "ENSG00000110092.4"]

# convert the dataframe to wide format from long format so genes are column headers for count values
two_genes_wide <- tidyr::pivot_wider(two_genes, names_from = X, values_from = Count)

# Create scatterplot of comparing two selected genes
gene_scatter <- ggplot(two_genes_wide, aes(x = ENSG00000184292.7, y = ENSG00000110092.4)) +
  # set color and size of points in scatterplot
  geom_point(color = "darkorchid4", size = 0.5) +
  # add title to plot and axes
  labs(title = "Comparison of Counts for Gene TACSTD2 vs Gene CCND1",
        x = "Gene TACSTD2 Count",
        y = "Gene CCND1 Count") +
  theme_minimal()

print(gene_scatter)
```

Comparison of Counts for Gene TACSTD2 vs Gene CCND1



Create Violin Plot

```
# read in metadata file
metadata <- read.csv("/Users/cassieschool/Documents/Foundations in Data Science/Final Project/Data/metadata.csv")

# subset metadata to only include one covariate of interest
covariate_metadata <- subset(metadata, select = c(barcode, paper_pathologic_stage))

# convert barcode values in metadata to have . instead of - so they can match up to data in count file
covariate_metadata$barcode <- gsub("-", ".", covariate_metadata$barcode)

# join count data and metadata by barcode/sample so gene counts can be compared to chosen covariate
gene_metadata <- merge(one_gene, covariate_metadata[, c("barcode", "paper_pathologic_stage")], by = "barcode")

# generate violin plot for distribution of count data based on covariate
# omitted NA values in plotting
gene_violin <- ggplot(na.omit(gene_metadata), aes(x = paper_pathologic_stage, y = Count, fill = paper_pathologic_stage)) +
  geom_violin() +
  # add title to plot and axes
  labs(title = "Distribution of Count of Gene TACSTD2 for Disease Pathologic Stage",
       x = "Disease Pathologic Stage",
       y = "Gene TACSTD2 Count") +
  # set plot color scheme
  scale_fill_brewer(palette = "Blues")

print(gene_violin)
```

